

PROJECT SYNOPSIS
ON
TrustBridge: An AI-Assisted, Blockchain-Enforced
Crowdfunding Platform with Milestone-Based
Escrow and Risk Assessment

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in

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JSS ACADEMY OF TECHNICAL EDUCATION,NOIDA

DR. APJ ABDUL KALAM TECHNICAL UNIVERSITY

UTTAR PRADESH, LUCKNOW

(Formerly Uttar Pradesh Technical University, Lucknow)

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PO3: Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO4: Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5: Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO6: The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO7: Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO8: Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO9: Individual and teamwork: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO10: Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and

receive clear instructions.

PO11: Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO12: Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM EDUCATIONAL OUTCOMES (PEOs)

PEO1: To apply computational skills necessary to analyze, formulate and solve engineering problems.

PEO2: To establish a entrepreneurs, and work in interdisciplinary research and development organizations as an individual or in a team.

PEO3: To inculcate ethical values and leadership qualities in students to have a successful career.

PEO4: To develop analytical thinking that helps them to comprehend and solve real-world problems and inherit the attitude of lifelong learning for pursuing higher education.

PROGRAM SPECIFIC OUTCOMES (PSOs)

PSO1: Acquiring in depth knowledge of theoretical foundations and issues in Computer Science to induce learning abilities for developing computational skills.

PSO2: Ability to analyse, design, develop, test and manage complex software system and applications using advanced tools and techniques.

Course Outcomes(COs)

C410.1: Identify, formulate, design and analyze a research based/web based problem.

C410.2: Communicate effectively in verbal and written form

C410.3: Apply appropriate computing, and engineering skills for obtaining solution to the formulated problem within a stipulated time.

C410.4: Work effectively as a part of team in multi-disciplinary areas.

C410.5: Consolidate the final outcome in the form of a publication.

CO-PO-PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
C410.1	3	3	3	3	2	3	3	3	3	3	2	3	3	3
C410.2	2	2	2	2	2	2	2	2	2	3	2	3	2	3
C410.3	3	3	3	3	3	3	2	3	3	3	3	3	3	3
C410.4	3	3	3	3	2	3	2	3	3	3	3	3	3	3
C410.5	3	3	3	3	3	3	3	3	3	3	3	3	3	3
C410	2.80	2.80	2.80	2.80	2.40	2.80	2.40	2.80	2.80	3.00	2.60	3.00	2.80	3.00

DECLARATION

We hereby declare that this synopsis submission is our own work and that, to the best of our knowledge and belief, it contains no material previously published or written by another person nor material which to a substantial extent has been accepted for the award of any other degree or diploma of the university or other institute of higher learning, except where due acknowledgment has been made in the text.

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CERTIFICATE

This is to certify that project synopsis report entitled "**TrustBridge: An AI-Assisted, Blockchain-Enforced Crowdfunding Platform with Milestone-Based Escrow and Risk Assessment**" which is submitted by Akshar Vikram (2300911530012), Akshat Mishra (2300911530013), and Aditya Kasoudhan (2300911530007) in partial fulfillment of the requirement for the award of degree B.Tech. in Department of Computer Science and Engineering-AIML of Dr. APJ Abdul Kalam Technical University, Uttar Pradesh, Lucknow is a record of the candidate's own work carried out by them under my supervision. The matter embodied in this report is original and has not been submitted for the award of any other degree.

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TrustBridge: An AI-Assisted, Blockchain-Enforced Crowdfunding Platform with Milestone-Based Escrow and Risk Assessment

ABSTRACT

TrustBridge is an AI-assisted, blockchain-enforced crowdfunding platform developed as a BTech final-year research prototype. Conventional crowdfunding platforms suffer from two fundamental limitations: contributors cannot reliably evaluate campaign quality or creator credibility prior to funding, and there is no programmable mechanism to enforce accountability after funding. TrustBridge addresses both limitations through an integrated architecture that combines Machine Learning (ML) campaign-success prediction, anomaly-based risk detection, Agentic AI explanation and evidence review, off-chain creator identity verification (KYC), and Solidity smart-contract escrow with milestone-based staged fund release.

At the time of campaign launch, a trained ML model — evaluated across Logistic Regression, Random Forest, and Gradient Boosting classifiers — estimates the probability of success using only launch-time features, thereby avoiding data leakage. An Isolation Forest model provides a complementary anomaly signal to flag statistically unusual campaigns. Four Agentic AI agents (Campaign Analyzer, Risk Analyst, Evidence Reviewer, and Explainer) synthesize these signals into human-readable assessments for contributors.

Contributor funds are held in a Solidity smart contract deployed on the Ethereum Sepolia testnet. The contract enforces a 20 ETH hard cap, releases an initial 20% development tranche upon goal attainment, and gates the remaining 80% across three subsequent milestone tranches. Each milestone requires the creator to submit evidence, which is analyzed by the Evidence Reviewer agent and then approved or rejected by an authorized human verifier. All financial events — contributions, refunds, tranche releases — are recorded on-chain and auditable via Etherscan.

The system is evaluated through four experiments: ML model comparison, explanation quality assessment, smart-contract correctness testing, and full end-to-end workflow validation. TrustBridge contributes a unified, research-ready prototype demonstrating that programmable escrow, AI-driven transparency, and human-verified milestones can meaningfully improve contributor trust and creator accountability in crowdfunding.

Keywords: Crowdfunding, Machine Learning, Agentic AI, Blockchain, Smart Contract, Escrow, Milestone Funding, Risk Assessment, Ethereum, Sepolia.

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LIST OF SYMBOLS/ ABBREVIATIONS

Symbol	Description
AI	Artificial Intelligence
API	Application Programming Interface
ABI	Application Binary Interface
dApp	Decentralised Application
ETH	Ether (Ethereum cryptocurrency unit)
KYC	Know Your Customer
LLM	Large Language Model
LR	Logistic Regression
ML	Machine Learning
MVP	Minimum Viable Product
NLP	Natural Language Processing
POC	Proof of Concept

Symbol	Description
REST	Representational State Transfer
RF	Random Forest
ROC-AUC	Receiver Operating Characteristic – Area Under Curve
RPC	Remote Procedure Call
SHAP	SHapley Additive exPlanations
TX	Transaction
UI/UX	User Interface / User Experience

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CHAPTER 1

INTRODUCTION

Crowdfunding has emerged as a widely adopted mechanism for early-stage project financing, enabling creators to raise capital directly from a global community of contributors. Platforms such as Kickstarter and Indiegogo have collectively facilitated billions of dollars in funding; however, they are fundamentally centralised systems that provide contributors with limited tools to assess campaign quality before committing funds, and no programmable mechanism to enforce creator accountability after funding is received.

Two structural problems persist in existing crowdfunding models. First, contributors rely almost entirely on self-reported campaign information — descriptions, images, and reward tiers — with no objective, data-driven signal about the likelihood of a campaign succeeding or the credibility of the creator behind it. Second, once funds are released to a creator, the platform has no enforceable obligation to ensure milestones are met; contributors have no recourse if a project stalls or fails to deliver.

TrustBridge addresses both problems through an integrated architecture that applies three complementary technologies to the crowdfunding context:

1. **Machine Learning and Agentic AI** — to assess, explain, and communicate campaign risk and success probability to contributors before they commit funds.
2. **Smart-Contract Escrow** — to enforce programmable financial rules: a hard cap, staged release tranches, and automatic refunds — without relying on platform discretion.
3. **Human-Verified Milestones** — to gate later funding tranches behind evidence of real-world progress, reviewed by AI and approved by an authorised verifier, thereby maintaining accountability through the project lifecycle.

The core principle governing TrustBridge is: *AI helps users decide. Humans verify real-world milestones. Smart contracts control predefined financial rules. Blockchain records the*

result.

TrustBridge is built on the Ethereum Sepolia testnet using Solidity smart contracts, a React.js and Tailwind CSS frontend with MetaMask wallet integration, a Python Flask backend, and a scikit-learn ML pipeline. Creator identity is verified off-chain through a mock KYC module, ensuring that no personally identifiable information is stored on the blockchain.

This synopsis describes the problem statement, objectives, proposed architecture, ML methodology, Agentic AI design, blockchain implementation, related work, hardware and software requirements, predicted timeline, and expected outcomes for the TrustBridge project.

CHAPTER 2

MOTIVATION

The motivation for TrustBridge arises from three observable gaps in the current crowdfunding landscape.

Gap 1 — Contributor Information Asymmetry. Contributors to crowdfunding campaigns face a fundamental information asymmetry: they must evaluate campaign quality based entirely on creator-supplied material, without any independent, data-driven assessment. Research by Elitzur et al. (2024) and Ardakani et al. (2024) demonstrates that ML models trained on historical crowdfunding data can predict campaign success with meaningful accuracy using only information available at launch time. Yet no mainstream crowdfunding platform exposes such a signal to contributors. TrustBridge closes this gap by computing and displaying an ML-derived success probability and an AI-generated risk explanation on every campaign page.

Gap 2 — Lack of Programmable Post-Funding Accountability. In conventional crowdfunding, once a campaign is successfully funded, the release of money is governed by platform policy — not by code. A creator who fails to deliver has, in practice, already received the funds. Smart-contract-based crowdfunding research (reviewed in Section 5) demonstrates that blockchain escrow can enforce predefined release conditions without relying on platform intermediaries. TrustBridge implements a four-tranche milestone escrow in Solidity: an initial 20% development tranche is released upon goal attainment, and the remaining 80% is released in three subsequent tranches only after an authorised verifier approves the corresponding milestone evidence. If a milestone is rejected, the contract executes a predefined refund rule, returning the unspent escrow to contributors.

Gap 3 — No Integrated Architecture Combining All Three Layers. Individually, ML crowdfunding prediction, blockchain escrow, and explainable AI have each been studied in the literature. What has not been evaluated is an integrated system in which these three

layers operate together: ML and Agentic AI at the assessment stage, smart-contract escrow at the funding stage, and AI-plus-human verification at the milestone stage. TrustBridge is motivated by the hypothesis that this integration produces a qualitatively different — and more trustworthy — crowdfunding experience than any single layer can provide alone.

Beyond the academic motivation, the team is personally motivated by the practical problem of trust in early-stage technology funding in India, where contributors increasingly encounter projects that are either misrepresented or fail to deliver, with no recourse available.

CHAPTER 3

OBJECTIVES

The primary objective of TrustBridge is to design, implement, and evaluate an integrated crowdfunding platform that combines ML-based campaign assessment, Agentic AI explanation, off-chain creator verification, and blockchain-enforced milestone escrow.

The specific objectives are as follows:

1. **Campaign Success Prediction** — Train and evaluate ML classifiers (Logistic Regression, Random Forest, Gradient Boosting) to predict crowdfunding campaign success using only launch-time features, reporting precision, recall, F1, ROC-AUC, and Brier score for each model.
2. **Risk and Anomaly Detection** — Apply an Isolation Forest anomaly detection model to identify statistically unusual campaigns and return a calibrated risk signal (LOW / MEDIUM / HIGH) accompanied by an explicit disclaimer that it is not a fraud verdict.
3. **Agentic AI Assessment and Explanation** — Implement four AI agents — Campaign Analyzer, Risk Analyst, Evidence Reviewer, and Explainer — that synthesize ML outputs, campaign metadata, and milestone evidence into structured, human-readable assessments using a large language model (LLM) backend.
4. **Creator Identity Verification** — Implement an off-chain mock KYC module that assigns creators a Verified / Pending / Rejected status, ensuring that no raw identity document is stored on the blockchain.
5. **Smart-Contract Escrow with Hard Cap** — Deploy a Solidity smart contract on the Ethereum Sepolia testnet that (a) enforces a 20 ETH hard cap, automatically refunding excess contributions; (b) holds all contributor funds in escrow; (c) releases an initial 20% tranche upon goal attainment; and (d) gates subsequent tranches behind approved milestones.

6. **Milestone-Based Staged Funding** — Implement a four-milestone release schedule (20%, 25%, 25%, 30%) in which each tranche is released only after the creator submits evidence, the Evidence Reviewer agent analyzes it, and an authorised human verifier approves or rejects the milestone.
7. **Contributor-Facing Transparency** — Present contributors with a unified campaign detail page showing creator verification status, ML success probability, AI risk explanation, milestone tracker, and on-chain transaction history — all in a single interface.
8. **Research Evaluation** — Conduct four structured experiments: (A) ML model comparison, (B) explanation quality assessment, (C) smart-contract correctness and edge-case testing, and (D) full end-to-end workflow validation, to produce results suitable for a research paper.

CHAPTER 4

SCOPE OF THE PROJECT

TrustBridge is scoped as an academic MVP (Minimum Viable Product) and research prototype targeting the Ethereum Sepolia testnet. The scope is defined along three dimensions: the proposed system architecture, what the system will include, and what it explicitly excludes.

4.1 Proposed System Architecture

TrustBridge is organised as a five-layer architecture in which the advisory intelligence layer and the financial-control layer are deliberately kept separate. The Application Layer (Flask + scikit-learn + Agentic AI) computes risk and success signals but never holds funds or private keys, while the Smart Contract Layer holds all contributor funds in escrow and executes only predefined, human-authorised release rules. Figure 4.1 shows the layered decomposition and the direction of data flow between layers.

The separation of concerns is the key design principle: advisory intelligence informs human decisions, and only a smart contract — triggered by an authorised human verifier — moves money.

4.2 End-to-End Workflow

Figure 4.2 summarises the operational sequence encoded by the core principle *Verify* → *Assess* → *Decide* → *Escrow* → *Build* → *Prove* → *Release/Refund* → *Audit*.

4.3 In Scope

- A fully functional web application with nine frontend pages: Home/Explore, Campaign Details, Create Campaign, Creator Dashboard, Contributor Dashboard, Mile-

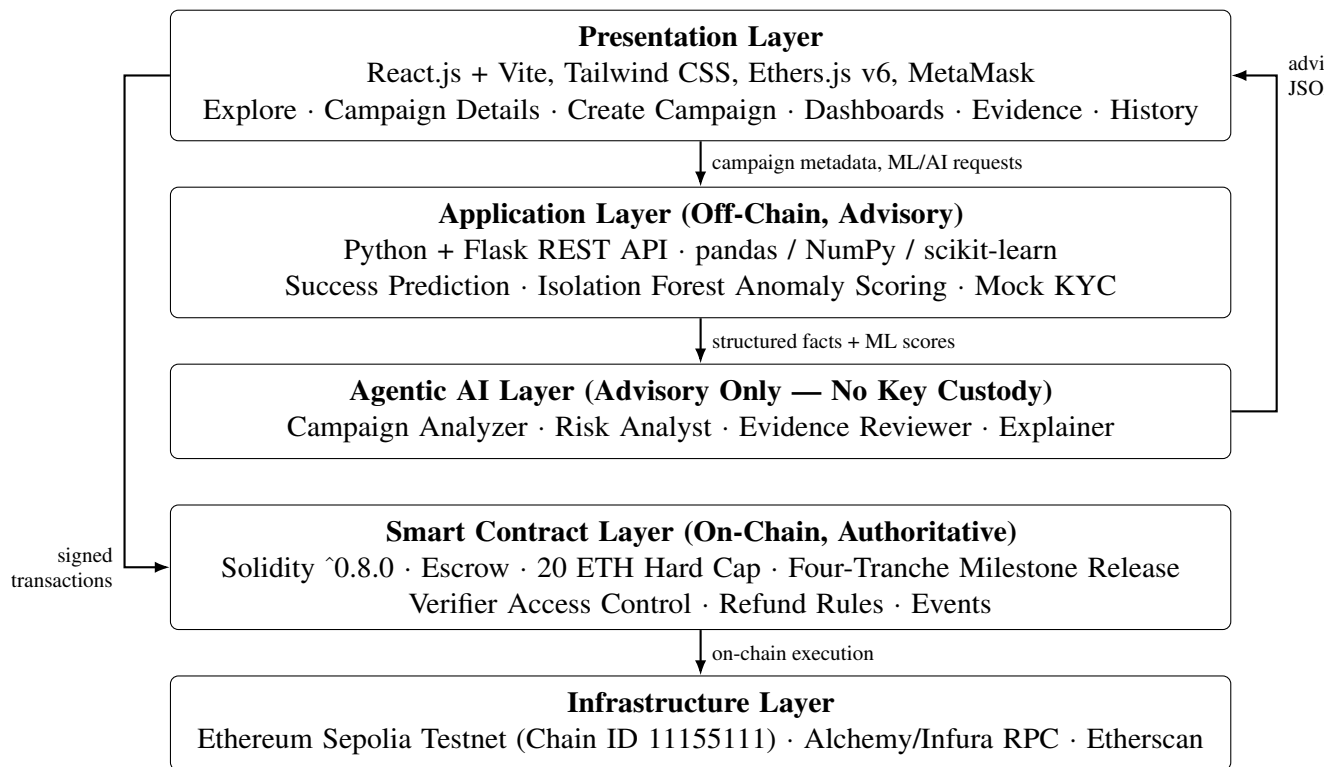


Figure 4.1: TrustBridge Five-Layer System Architecture

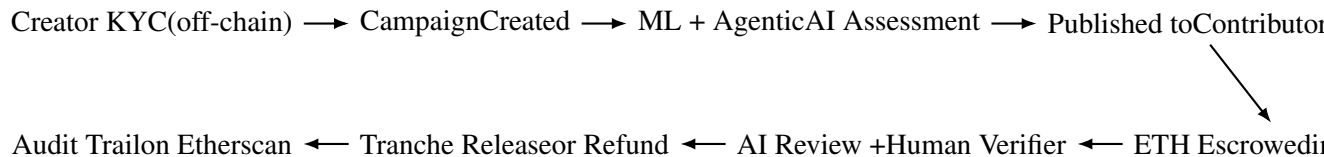


Figure 4.2: TrustBridge End-to-End Funding and Verification Workflow

stone/Evidence, Wallet, AI Risk Assessment, and Transaction History.

- A Python Flask backend exposing REST APIs for campaign metadata, ML prediction, anomaly scoring, Agentic AI analysis, and KYC status.
- A scikit-learn ML pipeline trained on a publicly available historical crowdfunding dataset (e.g., Kickstarter data from Kaggle), evaluated across three model families.
- An Isolation Forest anomaly detection module returning a risk signal per campaign.
- Four Agentic AI agents implemented in Python using an LLM API (NVIDIA Nemotron or equivalent), returning structured JSON outputs.
- A mock KYC module that simulates identity verification without storing real identity documents.

- A Solidity smart contract deployed to the Ethereum Sepolia testnet enforcing: hard cap (20 ETH), escrow, four-tranche milestone release, partial refunds, and access control.
- MetaMask wallet integration via Ethers.js v6.
- End-to-end testing covering at least 50 transactions on Sepolia, with all results auditable on Etherscan.
- A research paper draft structured for potential submission to an academic venue.

4.4 Out of Scope

- Production-scale deployment or mainnet Ethereum usage.
- Guaranteed fraud prevention or investment advice.
- Storage of real KYC documents (Aadhaar, PAN) on-chain or in any database.
- Automatic physical-world milestone verification without human involvement.
- Direct LLM control of financial transactions or private-key management.
- Support for blockchains other than Ethereum Sepolia.
- Legal compliance for a production crowdfunding product.

4.5 System Boundary

The system boundary of TrustBridge encompasses: (a) the React frontend running in the contributor/creator browser; (b) the Flask backend and ML pipeline running on a local or cloud server; (c) the Agentic AI agents calling an external LLM API; (d) the Solidity smart contract running on the Sepolia testnet; and (e) the off-chain SQLite database storing campaign metadata, KYC status, and AI analysis outputs. The blockchain itself and the MetaMask wallet are external dependencies outside the system boundary.

CHAPTER 5

RELATED WORK

A substantial body of literature exists across the individual components of TrustBridge. This chapter reviews the most directly relevant work and identifies the research gap that TrustBridge addresses.

5.1 Crowdfunding Success Prediction

Elitzur et al. (2024) conducted a comprehensive study of ML methods for crowdfunding success prediction, establishing that classification models trained on campaign metadata can achieve meaningful predictive accuracy. Their work provides the methodological baseline for TrustBridge’s ML module. Ardakani et al. (2024) extended this line of research by incorporating both numerical campaign features and textual campaign descriptions using NLP techniques, demonstrating that combined feature sets outperform numerical-only models. Feng et al. (2024) contributed feature-selection methods for compact, interpretable crowdfunding prediction models, informing TrustBridge’s feature engineering strategy.

5.2 Crowdfunding Fraud and Risk Detection

Perez et al. (2020), in their paper “I Call BS: Fraud Detection in Crowdfunding Campaigns,” applied ML to identify fraudulent or misleading campaigns using linguistic and metadata signals. This work is directly relevant to TrustBridge’s anomaly detection module. Their finding that success prediction and fraud detection are distinct tasks — requiring separate models — motivates TrustBridge’s architectural decision to implement a separate Isolation Forest risk module alongside the success prediction classifier.

5.3 Explainable ML in Crowdfunding

Gafrej et al. (2026) applied SHAP (SHapley Additive exPlanations) to explain ML predictions for technology reward crowdfunding campaigns, demonstrating that contributors respond more positively to predictions when accompanied by explanations. This work motivates TrustBridge’s Agentic AI explanation layer and constitutes empirical support for Experiment B (explanation quality assessment).

5.4 Investor and Contributor Dynamics

Porro et al. (2024) studied temporal dynamics in equity and lending crowdfunding using ML, finding that contributor behaviour evolves as a campaign progresses. While TrustBridge focuses on launch-time prediction, this work suggests a future direction of dynamic risk updating across the campaign lifecycle.

5.5 Blockchain-Based Crowdfunding

Recent research has examined blockchain as an infrastructure for crowdfunding, focusing on smart-contract escrow, fund traceability, and decentralised governance. Studies have demonstrated that Ethereum smart contracts can reliably enforce contribution limits, automate refunds, and release funds based on predefined conditions. TrustBridge builds directly on this body of work by implementing a production-quality escrow contract on Sepolia.

5.6 Research Gap

Individually, ML crowdfunding prediction, blockchain escrow, and explainable AI are well-studied. However, no prior work — to the best of the authors’ knowledge — has proposed and evaluated an integrated architecture that combines all four components: (1) ML success prediction and anomaly detection at the assessment stage, (2) Agentic AI explanation and evidence review across the campaign lifecycle, (3) off-chain creator identity verification, and (4) programmable milestone-based escrow enforced by a Solidity smart contract. TrustBridge addresses this gap and evaluates the integrated architecture empirically.

5.7 Conclusion

The related work establishes strong individual foundations for each component of TrustBridge but confirms the absence of an integrated, empirically evaluated system of the kind TrustBridge proposes. The research contribution is the integration and evaluation, not any single component in isolation.

5.8 Future Scope

Future work may extend TrustBridge in several directions: (a) dynamic risk scoring that updates as campaign contributions progress; (b) multi-signature verifier approval for milestone decisions; (c) deployment to Ethereum mainnet with real KYC compliance; (d) NLP-based analysis of creator communication patterns across the campaign lifecycle; and (e) a decentralised autonomous organisation (DAO) governance layer for verifier selection.

CHAPTER 6

HARDWARE & SOFTWARE REQUIREMENTS

6.1 Hardware Requirements

Component	Minimum Specification
Processor	Intel Core i5 (8th Gen) or equivalent, 2.0 GHz
RAM	8 GB DDR4
Storage	20 GB free disk space (SSD preferred)
Display	1366 × 768 resolution
Internet	Stable broadband connection (for Sepolia RPC and LLM API)
Browser	Google Chrome 110+ or Firefox 110+ (for MetaMask)

Table 6.1: Hardware Requirements for TrustBridge Development and Testing

6.2 Software Requirements

6.2.1 Frontend

- React.js 18.x with Vite build tool
- Tailwind CSS 3.x
- Ethers.js v6 (blockchain interaction)
- MetaMask browser extension (wallet)
- Axios (HTTP client for Flask API)
- React Router v6 (client-side routing)
- Recharts (campaign progress charts)

6.2.2 Backend

- Python 3.10+
- Flask 3.x (REST API framework)
- Flask-CORS (cross-origin resource sharing)
- SQLite (off-chain campaign and KYC metadata)
- python-dotenv (environment variable management)

6.2.3 Machine Learning

- pandas, NumPy (data preprocessing)
- scikit-learn (Logistic Regression, Random Forest, Isolation Forest)
- XGBoost / Gradient Boosting (optional comparison model)
- joblib (model serialisation and loading)
- Historical crowdfunding dataset (Kaggle Kickstarter dataset)

6.2.4 Agentic AI

- NVIDIA Nemotron or equivalent LLM (via API)
- Python orchestration layer (custom agent classes)
- requests (LLM API HTTP calls)

6.2.5 Blockchain

- Solidity ^0.8.0 (smart contract language)
- Remix IDE (contract development and deployment)
- Ethereum Sepolia testnet (Chain ID: 11155111)
- Alchemy or Infura (Sepolia RPC provider)

- Etherscan Sepolia (transaction verification)
- MetaMask (testnet wallet)
- Test ETH via Sepolia faucet (sepoliafaucet.com)

6.2.6 Development Tools

- Visual Studio Code / Antigravity IDE (code editor)
- Git and GitHub (version control)
- Node.js 18+ and npm (frontend package management)
- Postman (API testing)
- Google Sheets or Excel (test data recording)

6.2.7 Operating System

- Windows 10/11, macOS 12+, or Ubuntu 20.04+ (any of these is sufficient)

CHAPTER 7

PREDICTED TIMELINE FOR THE PROJECT

The project is planned over 12 weeks following the initial synopsis approval. The timeline is divided into two phases: a 4-week Proof of Concept (POC) phase and an 8-week Full System phase.

Week	Phase	Activity	Deliverable
1	POC	Write and deploy TrustBridgePOC.sol to Sepolia	Deployed contract, ABI, Etherscan link
2	POC	Build minimal React frontend; MetaMask integration	Working contribute flow
3	POC	Execute 9 manual test scenarios; record gas costs	Test results spreadsheet
4	POC	Write POC test report; prepare viva summary	POC documentation
5–6	Full	Expand contract: dynamic campaigns, multi-milestone, verifier	TrustBridge.sol deployed
7–8	Full	Flask backend + ML pipeline (3 models, all metrics)	Trained .pkl models, API endpoints
8–9	Full	Agentic AI agents + mock KYC module	4 agent classes, KYC API
9–10	Full	Full React UI (9 pages) with Tailwind CSS	Complete frontend
10–11	Full	Integration testing; 50+ Sepolia transactions	Test report, Etherscan links
12	Full	Final documentation, research paper draft, viva preparation	Synopsis, paper, presentation

Table 7.1: Predicted 12-Week Project Timeline

Milestone Summary

- **End of Week 4 (POC complete):** Smart contract deployed and tested; minimal React frontend working; 9 test scenarios executed; POC documentation ready.
- **End of Week 8 (Backend complete):** Full Solidity contract live on Sepolia; Flask REST APIs operational; ML models trained and evaluated.
- **End of Week 11 (Integration complete):** All 9 frontend pages complete; end-to-end flow validated with 50+ transactions; Agentic AI and KYC integrated.

- **End of Week 12 (Submission ready):** Final documentation, research paper draft, and viva presentation prepared.

CHAPTER 8

CONCLUSION

TrustBridge proposes and will evaluate an integrated crowdfunding architecture that addresses two fundamental limitations of existing platforms: the inability of contributors to objectively assess campaign quality before funding, and the lack of programmable accountability mechanisms after funding.

By combining ML-based success prediction, anomaly-driven risk detection, Agentic AI explanation and evidence review, off-chain creator verification, and Solidity smart-contract escrow with milestone-based staged fund release, TrustBridge creates a system in which transparency is data-driven, accountability is code-enforced, and trust is built into the protocol rather than assumed from the platform.

The project is grounded in a substantive body of related work across ML crowdfunding prediction, blockchain escrow, and explainable AI. Its research contribution lies in the integration and empirical evaluation of all these components in a single, end-to-end system — a gap confirmed by the literature review in Chapter 5.

The 12-week implementation plan, beginning with a 4-week POC phase on the Ethereum Sepolia testnet, is designed to progressively validate each system component before full integration. The POC phase alone will produce on-chain evidence of correct hard-cap enforcement, partial contribution refunds, milestone-gated withdrawal, and automatic refund execution — demonstrating the core TrustBridge value proposition with verifiable data.

Upon completion, TrustBridge will be a research-ready prototype suitable for academic evaluation, supervisor demonstration, and as the basis of a conference or journal paper contribution. It is expected to demonstrate that an integrated AI-plus-blockchain architecture can meaningfully improve both the pre-funding decision quality and the post-funding accountability of crowdfunding campaigns.

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APPENDIX A

PLAGIARISM REPORT

A.1 Plagiarism Report

[Attach plagiarism report from Turnitin or iThenticate here. The synopsis content is original academic work. All references to prior literature are cited in Chapter 5 (Related Work) and in the References section.]

A.2 AI Content Report

[Attach AI content detection report here if required by institution. The project concept, architecture, and technical decisions are the original work of the team members under the supervision of Ms. Upasana.]